Chapter 59. Extubation

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Extubation: Introduction

It is easy to merge decisions about extubation with decisions about weaning in everyday practice. Indeed, much patient mismanagement is caused by conflating these two subjects. But the conflation is not confined to clinicians. Many researchers have also merged the two subjects, such as using weaning predictors to predict reintubation in a patient who has already tolerated a weaning trial. The result is scientific confusion.

When a patient tolerates a weaning trial without distress, a clinician feels reasonably confident that the patient will be able to sustain spontaneous ventilation after extubation. But this is not the only consideration. The clinician also has to consider whether the patient will be able to maintain a patent upper airway after extubation.

Removal of an endotracheal tube is typically performed under controlled conditions. The patient has satisfactorily tolerated a weaning trial. Enteral feeding is temporally withheld for approximately 4 hours. The patient is usually positioned in a sitting posture. The endotracheal tube, mouth, and upper airway are suctioned, paying attention to the collection of secretions above an inflated cuff. Some clinicians recommend keeping a suction catheter in place (aiming for the catheter to barely protrude from the distal end of the endotracheal tube) as the cuff is deflated; this step is taken in an attempt to capture any secretions sitting on top of an inflated cuff, which might fall into the airway after deflating the cuff. Some clinicians forcefully inflate the lungs with an Ambu Bag immediately before pulling out the endotracheal tube, hoping that the larger than usual ensuing exhalation will push secretions upward and outward. After removal of the endotracheal tube, the patient is given supplemental oxygen, titrated to oxygen saturation (S_{O_2}), being particularly cautious with a patient who is at risk of carbon dioxide retention. Patients may have impaired airway protection reflexes immediately after extubation. If speech is impaired for more than 24 hours, indirect laryngoscopy should be undertaken to assess vocal cord function. Oral intake should be delayed in patients who have been intubated for a prolonged period.

In the hours following extubation, patients are carefully monitored for ability to protect the upper airway and sustain ventilation. Most patients will display progressive improvement, allowing the discontinuation of supplemental oxygen and ultimate discharge from the intensive care unit (ICU).

Postextubation Distress

Between 2%[1,2] and 30%[3–6] of patients experience respiratory distress in the postextubation period (Table 59-1). Many, but not all, require reinsertion of the endotracheal tube and mechanical ventilation. These patients are commonly classified as *extubation failures,* a term popularized by Demling et al.[7] These investigators defined extubation failure as the need for reintubation within 7 days. Unfortunately, the meaning of extubation failure varies among authors, leading to scientific confusion. Even when authors employ it as a synonym for reintubation, the period under study varies—within 24, 48, or 72 hours, or as long as 7 days.

Table 59-1: Frequency of Reintubation and Mortality

Authors	Number of Patients	Percent Reintubated	Percent Mortality	Time Frame
Tahvanainen et al[73]	47	19.1	22.2	—
DeHaven et al[74]	48	6.3	NR	—
Demling et al[7]	400	4.4	40	7 days
Demling et al[7]	299	3.3	10	7 days
Krieger et al[75]	269	10.4	NR	—
Mohsenifar et al[76]	29	14.3	NR	24 hours
Sassoon et al[77]	40	12.5	NR	48 hours
Brochard et al[78]	109	11	NR	48 hours
Lee et al[79]	52	17	33.3	NR
Capdevila et al[80]	67	17.9	NR	48 hours
Esteban et al[81]	530	15.6	NR	48 hours
Torres et al[29]	170	23.5	35	—
Chatila et al[82]	100	9.5		<24 hours
Dojat et al[3]	38	29.4	40	48 hours
Ely et al[83]	300	3.7	NR	48 hours
Leitch et al[1]	163	1.8	NR	<24 hours
Miller et al[35]	88	17	NR	NR
Epstein et al[12]	289	14.5	42.5	—
Esteban et al[28]	484	18.6	27	48 hours
Jacob et al[84]	183	4.5	NR	24 hours
Kollef et al[85]	357	11.5	NR	NR
Vallverdu et al[86]	217	15.5	NR	48 hours
Esteban et al[15]	526	13.5	32.8	48 hours

Authors	Number of Patients	Percent Reintubated	Percent Mortality	Time Frame
Zeggwagh et al[33]	101	37	NR	48 hours
Coplin et al[45]	136	17.6	NR	NR
Koh et al[87]	36	19	NR	48 hours
Maldonado et al[5]	24	26.7	NR	24 hours
Khamiees et al[43]	91	12.8	NR	72 hours
Namen et al[13]	100	16	NR	NR
Cohen et al[4]	35	28.6	10	48 hours
De Bast et al[22]	76	18.4	NR	24 hours
Perren et al[88]	98	6.7	33	48 hours
Smina et al[16]	95	11.3	—	72 hours
Conti et al[2]	92	1.7	NR	48 hours
Fernandez et al[89]	130	18	NR	48 hours
Francois et al[21]	343	8	0.3	24 hours
Thille et al[31]	168	15.5	50%	72 hours

Abbreviation: NR, not reported.

Extubation failure is most often defined as the need for reintubation. The corollary of this definition is that all patients who do not require reintubation should be classified as extubation successes no matter how much difficulty they experience. A number of patients develop stridor after extubation, which resolves with inhalation of racemic epinephrine or other therapy without requiring reintubation. If extubation failure is defined as reintubation, such patients should be excluded. But if researchers are investigating the development of respiratory distress after extubation, all patients with significant postextubation stridor should be included. Much confusion could be avoided if researchers used the term *reintubation* when that is their sole criterion for extubation failure. If researchers use the term *extubation failure,* it seems logical to assume that their study population includes some patients with postextubation distress who did not require reintubation. In many reports, however, researchers do not make it clear which approach they are adopting—whether (or not) their category of extubation failure included some patients who developed distress after extubation but who did not require reintubation.

Investigators have variably classified patients who required noninvasive ventilation (NIV) after extubation as satisfying or not satisfying the definition of extubation failure. De Lassence et al[8] specified that they excluded patients who were managed by NIV in a cohort of extubation failure patients. Maldonado et al[5] included patients managed by NIV among their group of extubation failures, as did Haberthur et al[9] and Jiang et al.[10] Moreover, Haberthur et al[9] extended the term *extubation failure* to include patients experiencing unjustified delay in extubation with one particular weaning technique if that patient tolerated extubation after being switched to an alternative weaning technique.

If patients die from a cardiorespiratory cause without being reintubated, it seems unscientific to classify them as extubation successes. Some authors[11] have specified that their definition of extubation failure included both the need for reintubation or unexpected death within 72 hours. Most authors, however, do not address this issue. Some extubated patients refuse reintubation (as part of a decision for withdrawal of life support) and die. How should these patients be classified? Demling et al[7] classified fatal outcomes as extubation failures, Epstein et al[12] excluded such patients from their group of extubation failures, and Namen et al[13] classified such patients (who died) as extubation successes.

How should patients who experience unplanned extubation followed by reintubation be classified? In a group of forty-two extubation failures, Epstein et al[12] included four unplanned extubations (who required reintubation within 72 hours) because these occurred while a weaning trial was in progress, but excluded sixteen other unplanned extubations (requiring immediate extubation) because these did not occur during weaning trials. If patients are extubated because of a defective endotracheal tube and then reintubated, how should they be classified? Epstein and Ciubotaru[14] excluded such patients, but many authors do not state clearly how such patients are classified.

Some patients may be inappropriately reintubated because of poor clinical judgment. What criteria should be set for judging a reintubation as appropriate? Epstein and Ciubotaru[14] listed the following criteria: increase in arterial carbon dioxide tension (P_{CO_2}) greater than 10 mm Hg and decrease in pH of 0.10; arterial oxygen tension (P_{O_2}) less than 60 mm Hg or less than 90% with an inspired oxygen concentration (F_{IO_2}) greater than 0.50; signs of increased work of breathing (high respiratory rate, accessory muscle use, or paradoxical breathing); and inability to protect the airway (secondary to upper airway obstruction or excess secretions). Many of these criteria are similar to those used for defining weaning failure, but it is more difficult to ensure their rigorous and consistent application as criteria for reintubation. Weaning failure typically occurs under controlled conditions, usually within 1 hour of starting a weaning trial. Reintubation for postextubation distress may not occur until many hours after extubation, and the listing of satisfied criteria may not be entered on a data form until hours after the event.

Causes and Pathophysiology of Postextubation Distress

The listed indications for reintubation vary considerably from study to study. Of these, postextubation stridor has attracted the most attention.

Postextubation Upper Airway Obstruction

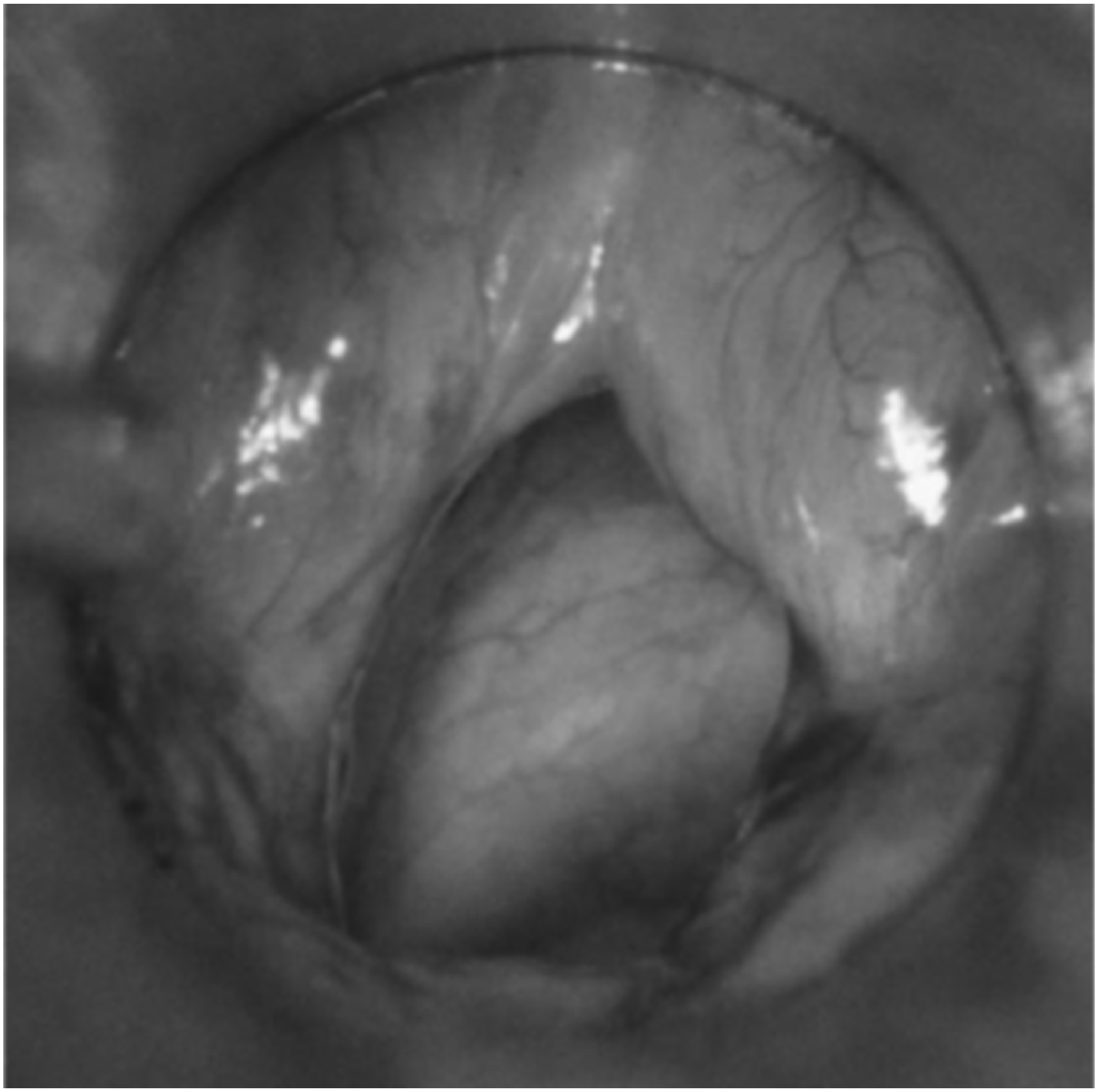
A number of investigators have reported that upper airway obstruction accounts for approximately 15% of patients requiring reintubation (15% in the study of Epstein and Ciubotaru;[14] 14.7% in that of Esteban et al;[15] and 15.4% in that of Smina et al[16]). These investigators, however, did not report what proportion of patients who developed clinical manifestations of upper airway obstruction did not require reintubation. Upper airway obstruction may result from edema of the subglottic area or the vocal cords, reflex closure of the vocal cords (laryngospasm), or compromise of the tracheal lumen (tracheomalacia or compression by a hematoma) (Figs. 59-1 and 59-2). The laryngeal edema that occurs in intubated patients is believed to arise from direct mechanical trauma to the larynx by the endotracheal tube. Almost every patient intubated for 4 days or more develops laryngeal edema and mucosal ulcerations, usually located posterior to the level of the vocal cords, where the tube exerts the highest pressure.[17] Laryngeal edema is usually transient and self-limiting, and most of the lesions resolve within 1 month.[18]

Figure 59-1

Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Mild postextubation laryngeal edema as seen with a rigid laryngoscope. (Used, with permission, from Antonaglia et al.[102])

Figure 59-2



Moderately severe postextubation laryngeal edema as seen with a rigid laryngoscope. (Used, with permission, from Antonaglia et al.[102])

Upper airway obstruction causes stridor only if the patient is capable of generating sufficient airflow; if airflow is insufficient, obstruction may cause hypercapnia, hypoxemia, or paradoxical breathing. Of 110 extubated patients, Sandhu et al[19] observed that thirteen (11.8%) developed stridor, but less than half of the patients with stridor (6/13) required reintubation (no patient required reintubation for any reason other than stridor in this series). Jaber et al[20] observed stridor in thirteen of 112 (11.6%) extubated patients, and nine of the thirteen required reintubation (only 2% [2/99] of patients without stridor required reintubation). Francois et al[21] undertook a randomized controlled trial of the effect of steroids on postextubation laryngeal edema in 761 critically ill patients requiring at least 36 hours of mechanical ventilation. Of the 343 patients in the placebo arm, seventy-six (22%) exhibited features of postextubation laryngeal edema. Twenty-six of the 343 extubated patients (8%) required reintubation, and the reintubation was linked to laryngeal edema in fourteen (54%[14/26]) of these patients. Forty-six percent of the patients with postextubation stridor or endoscopic laryngeal edema did not require reintubation, indicating that the development of stridor is not a particularly strong predictor of the need for reintubation.[17]

In the above studies, upper airway obstruction was typically diagnosed on the basis of clinical manifestations. De Bast et al[22] undertook fiber-optic examination before reintubation or directly inspected the glottis during reintubation to confirm the presence of laryngeal edema. Of seventy-six patients who had been intubated for at least 12 hours, fourteen required reintubation within 24 hours of extubation (reintubation rate, 18.4%). Of these fourteen patients, eight (57.1%) had laryngeal edema.

When upper airway obstruction occurs, it is typically manifested soon after extubation. Of the seventy-six patients who developed features of laryngeal edema in the study of Francois et al,[21] 47% (36/76) developed it within 5 minutes of extubation, 34% (26/76) developed it between 6 and 30 minutes after extubation, and 18% (14/76) developed it 30 minutes or longer after extubation.

Although laryngeal edema can develop as early as 6 hours after intubation,[22] many, but not all,[23] investigators have noted that the rate of postextubation stridor increases in proportion to the duration of ventilation. Jaber et al[20] observed that duration of intubation was longer in thirteen patients with stridor than in ninety-nine patients without stridor: 10.9 versus 5.5 days. Sandhu et al[19] likewise observed a longer duration of intubation in six patients who developed postextubation stridor than in ninety-seven patients who did not: 6.5 ± 1.9 versus 2.6 ± 2.6 days. Darmon et al[24] observed that stridor was more common among patients intubated for longer than 36 hours than among patients intubated for a shorter time: 7.2% (25/346) versus 0.9% (3/317).

Women are more susceptible to postextubation stridor than men, and the rate may vary with ethnicity. In a series from France, Darmon et al[24] observed stridor in 7.4% (20/284) of women and 2.1% (8/379) of men. In a series from Taiwan, Ho et al[23] observed stridor in 39% (7/18) of women and 17% (10/59) of men.

Other risk factors associated with the development of laryngeal edema include traumatic intubation, excessive tube size, excessive tube mobility secondary to insufficient fixation, a patient fighting against the tube or trying to speak, excessive pressure in the cuff, too frequent or too aggressive tracheal suctioning, occurrence of infections or hypotension, and the presence of a nasogastric tube that predisposes to gastroesophageal reflux.[22] It is also possible that a biochemical reaction between the tube material and the airway mucosa may cause laryngeal edema.[22] Compared with the ninety-nine patients without stridor, the thirteen patients who developed stridor in the series of Jaber et al[20] were more likely to have the following: a traumatic and/or difficult intubation (54% vs. 7%), a history of self-extubation (38% vs. 4%), a higher balloon cuff pressure (83 vs. 40 cm H_2O), a higher simplified acute physiology score (SAPS) II score (50 vs. 38), and a medical rather than a surgical reason for admission (46% vs. 18%).

Other Causes of Postextubation Distress

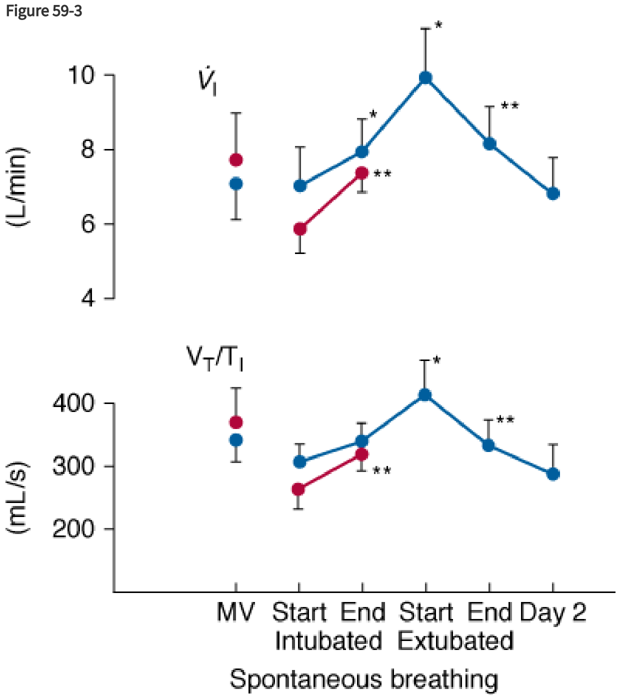
Conditions other than upper airway obstruction that cause postextubation distress vary from study to study. In a report on reasons for reintubation, Epstein and Ciubotaru[14] noted upper airway obstruction in 15%; other reasons for reintubation included respiratory failure (28%), congestive heart failure (23%), aspiration or excessive secretions (16%), encephalopathy (9%), and other conditions (8%). The frequency of a particular reason differs among studies. For example, cardiac failure accounted for 23% of the cases of Epstein and Ciubotaru[14] and 6.6% (4/61) of the cases of Esteban et al,[15] but none of the cases of Smina et al[16] or De Bast et al.[22] Because of the limited rigor of these studies, there is little point in attempting a more detailed analysis of the relative incidence of other causes of reintubation.

Pathophysiology

None of the studies of reasons for postextubation cardiorespiratory distress can be considered as studies of pathophysiologic mechanisms in the same sense as are studies of the pathophysiology of weaning failure. For studies of postextubation distress, investigators filled out case report forms. These forms constitute post hoc incident reports completed after some event. In many cases, investigators are making a best guess as to what might explain a patient's deterioration. In contrast, research into the pathophysiology of weaning failure is based on the simultaneous recording of several signals, starting before a weaning trial and continuing until after its completion. In this way, it is possible to understand the relative roles of control of breathing, respiratory muscle activity, derangements of lung and chest wall mechanics, gas exchange, and cardiovascular performance in weaning failure. Conducting similar types of studies to delineate the mechanisms of postextubation cardiorespiratory distress will be challenging.

The first challenge is instrumentation. The recording of swings in intrathoracic pressure, as reflected by esophageal pressure, is relatively easy. But on its own, esophageal pressure is of limited value. Derivation of most indices, such as airway resistance, compliance, and intrinsic positive end-expiratory pressure, require a simultaneous measurement of airflow. It is extremely difficult to obtain a meaningful measure of airflow or tidal volume in a recently extubated patient.[25] The use of a mouthpiece or face mask causes marked distortion of the breathing pattern. Inductive plethysmography provides a means for overcoming this problem. For example, Tobin et al[26] used this technique to study changes in breathing pattern in ten patients at the point of extubation (Fig. 59-3). During the first 15 minutes after extubation, both minute ventilation and mean inspiratory flow (a measure of respiratory motor output) increased, accompanied by a decrease in the degree of abdominal paradox. By the end of the first hour after extubation, respiratory drive and minute ventilation had returned to preextubation levels. No further change in breathing pattern was observed over the subsequent 24 hours. The investigators did not study any patients who developed postextubation distress. When inductive plethysmography is combined with esophageal pressure recordings, great care is required to ensure that the two signals are perfectly aligned. The smallest misalignment will cause major errors in estimates of intrinsic positive end-expiratory pressure and other measures of lung mechanics. A requirement not faced by researchers studying the pathophysiology of weaning failure is the need to record the development of laryngeal obstruction. As such, additional research instrumentation includes fiber-optic endoscopy.

Figure 59-3



Recordings of minute ventilation ($\dot{V}_I$) and mean inspiratory flow (V_T/T_I), a measure of respiratory motor output, during mechanical ventilation (*MV*), at the start and end of a T-tube trial, the first 15 minutes after extubation, 45 to 60 minutes after extubation, and 24 hours after extubation. Ten patients who tolerated the T-tube trial and were extubated are shown in *blue* and seven patients who failed the T-tube trial and were reconnected to the ventilator are shown in *red.* Bars represent ±1 standard error (SE); *, $p < 0.05$; **, $p < .01$ compared with the value during the immediately preceding time-block of spontaneous breathing. (Used, with permission, from Tobin et al.[26])

Perhaps an even greater challenge than the instrumentation is the timing. Weaning failure almost invariably occurs within the first hour of attempted spontaneous breathing. The time course for the development of postextubation cardiorespiratory distress extends over a longer span. In the study of Epstein and Ciubotaru,[14] for example, only 33% of reintubations occurred within the first 12 hours after extubation, and 42% occurred after 24 hours.

Consequences of Postextubation Distress

Many, but not all,[27] investigators have reported that mortality is many times higher in patients who require reintubation than in patients who tolerate extubation (Table 59-2). Three explanations have been offered to account for the increased mortality: complications associated with the act of reintubation itself; development of a new problem in the interval between extubation and reintubation; and the need for reintubation is simply serving as a marker for a poor prognosis.

Table 59-2: Association between Reintubation and Mortality

Reference	Mortality in Reintubated Patients	Mortality in Patients Tolerating Extubation
Daley et al[27]	8%	6.5%
Epstein et al[12]	43%	9.1%
Esteban et al[28]	27%	2.6%
Esteban et al[15]	33%	4.6%
Perren et al[88]	33%	3.6%
Thille et al[31]	50%	4.9%

Endotracheal intubation is typically performed under elective and controlled conditions. It is more challenging to perform intubation in a patient developing acute distress in the period after extubation. Complications have been reported to occur at the time of reintubation in 15%,[28] 18%,[15] and 28% of patients.[14] In a study of forty consecutive patients requiring reintubation (for any reason), Torres et al[29] reported that 47% (19/40) developed nosocomial pneumonia after reintubation, as compared with 10% of matched control patients (odds ratio [OR]: 5.9). In a study of 297 intubations performed under emergency conditions (in 238 adults), Schwartz et al[30] reported seven deaths (mortality: 2.4%) at the time of or within 30 minutes after intubation; five of the deaths were associated with a systolic blood pressure less than 90 mm Hg. In contrast to the experience of Torres et al,[29] only 4% of patients developed a radiographic infiltrate compatible with a new aspiration pneumonia.[30] In a study by Esteban et al,[15] mortality was no greater among the eleven patients who developed complications at the time of reintubation than in the remaining fifty patients (45.4% and 30.0%, respectively; $p = 0.53$). Based on the above considerations, it seems unlikely that the higher mortality in reintubated patients is a direct consequence of complications associated with the act of reintubation itself.

A second explanation is the development of a new problem during the interval between extubation and reintubation. In support of this possibility is the observation of Epstein and Cibotaru[14] that mortality increases in proportion to the time between extubation and reintubation: mortality of 69% in patients reintubated between 49 and 72 hours after extubation (17% of the group), 24% in patients reintubated in the first 12 hours after extubation (33% of the group), and 39% in patients reintubated between 13 and 24 hours after extubation (25% of the group).

The third explanation for higher mortality in reintubated patients is that reintubation is simply serving as a marker for a poor prognosis. Sicker patients are more likely to undergo reintubation. Epstein and Ciubotaru[14] have argued against this explanation. They note that reintubation continues to have a strong independent effect on mortality even after controlling for generalized severity of illness at weaning onset, comorbidity, age, and need for acute dialysis. It is, however, possible that the need for reintubation is measuring some additional aspect of disease severity not captured by the above variables.

Thille et al[31] have argued that reintubation has a direct and specific effect on patient outcome because it is frequently followed by marked clinical deterioration. They studied twenty-six patients who failed extubation and required reintubation within the subsequent 72 hours; 50% of the reintubated patients died in the ICU. At the time of extubation, the Sequential Organ Function Assessment (SOFA) score was not significantly different between the twenty-six patients who subsequently required reintubation and 142 patients who tolerated extubation. In the first 24 hours after extubation, the SOFA score increased significantly from 3.4 ± 2.9 to 4.7 ± 3.4 in the failed extubation group, mainly as a result of hemodynamic and respiratory deterioration. Patients who tolerated extubation showed an improvement in their SOFA score over the same interval, essentially because of removal of the ventilator. These data suggest that reintubation leads to adverse consequences, and it is not simply a marker for a poor prognosis.

Predictors of Postextubation Distress

Because reintubation causes serious complications in some patients, attempts are made to predict its likely occurrence. A number of physiologic variables have been evaluated for their ability to predict this likelihood. For some patients, the likelihood of reintubation is considered so high that a clinician may proceed to tracheotomy without first attempting extubation.

Ability to Sustain Spontaneous Ventilation

It is extremely uncommon to undertake planned extubation without first assessing a patient's ability to sustain spontaneous ventilation. This assessment typically consists of observing a patient breathing through a T-tube circuit or while assisted by a low level of pressure support or intermittent mandatory ventilation. A weaning trial serves primarily as an additional diagnostic test, with the aim of predicting whether a patient will develop distress after extubation and need reintubation. The predictive accuracy of a weaning trial as a diagnostic test has never been evaluated in a rigorous scientific manner.

A true-positive result of a T-tube trial is defined as a patient who tolerates the trial without distress, is then extubated, and does not require reintubation. The usual rate of reintubation is 15% to 20% (sometimes lower), but higher reintubation rates have been reported by some investigators: 23.5%,[29] 25%,[32] 26.7%,[5] 28.6%,[4] and 29.4%.[3] These false-positive test results mean that the positive-predictive value and specificity of passing a T-tube trial in predicting that a patient will not require reintubation is much less than 100%. To measure the false-negative rate would require extubating patients who fail a T-tube trial, and counting how many do not require reintubation. For obvious ethical reasons, we do not know this number. Given the natural caution of physicians, we can confidently assume that it is higher than 0%. As such, sensitivity and negative-predictive value will be less than 100%. It is no surprise that the ability of a T-tube trial to predict reintubation has a sensitivity and specificity of less than 100%; no diagnostic test is perfect. But a weaning trial is not solely used as a diagnostic test for predicting the likelihood of reintubation. The outcome of a weaning trial is also used as a reference standard against which the accuracy of weaning predictor tests are measured.

Weaning Predictor Tests

Several investigators have investigated the ability of weaning predictor tests to predict the development of distress after extubation. The question posed is along these lines: "Does frequency-to-tidal-volume ratio (f/V_T), or some other predictor test, measured before a T-tube trial, predict the likelihood of reintubation?" To answer this question with scientific validity, it is imperative that the investigators take clearly defined steps to ensure that clinicians are *not* taking the results of the T-tube trial into account when deciding whether to extubate the study patients. (In other words, a decision to extubate the patient must be taken before the T-tube trial, and must proceed even if the patient exhibits significant distress during the trial.) If researchers allow clinicians to use results of a T-tube trial (done after measurement of the weaning predictor test) when deciding whether or not to proceed with extubation, the researchers need a different experimental design because they are asking a different research question. The question is now, "In what instances do weaning predictor tests override the results of a subsequently undertaken T-tube trial?"

Before we discuss the findings of studies on the use of tests to predict the likelihood of postextubation distress, we ask the reader to undertake a simple thought experiment. You, as the patient's clinician, record f/V_T, and obtain a reading of 60. You then proceed to a T-tube trial. If the patient develops severe distress during the trial, would you extubate the patient? (Please exclude circumstances in which you believe that the internal diameter of the endotracheal tube is the main cause of distress.) We believe that most experienced clinicians will answer"no." Consider another scenario: A resident measures f/V_T in your patient, obtains a value of 120, and proceeds to a T-tube trial. If the patient tolerates the trial without significant distress, would you defer extubation? We believe most experienced clinicians would again answer "no," although they might monitor the postextubation period more closely than if the patient had an f/V_T reading of 90 before the trial.

For both of the preceding scenarios, we believe that few if any experienced clinicians would allow a measurement of f/V_T (made before a T-tube trial) to override a judgment based on how well a patient tolerates a T-tube trial. Given the results of the thought experiments, it makes little sense to undertake research studies of the accuracy of weaning predictor tests (measured before a weaning trial) to forecast a patient's likely need for reintubation (in a patient who passes a weaning trial). It makes even less sense when one considers that a clinician's action based on the patient's performance during the weaning trial will have inevitably muddied the experimental waters. Let us consider a patient in whom a weaning predictor test predicts a high likelihood of respiratory distress. A weaning trial is nevertheless undertaken. The patient develops distress, and so is not extubated. By excluding such patients from a study, the investigators are markedly underestimating the true-negative rate of the test for predicting distress after extubation (where the test predicts distress and the patient actually develops distress).

It is difficult to understand why so many investigators have undertaken this type of research. We suspect they have been seduced by affirmative answers to two subsidiary questions. “Do patients with satisfactory weaning predictors usually tolerate a spontaneous breathing trial?” Yes. “Do patients who tolerate a spontaneous breathing trial usually avoid reintubation?” Yes. It might seem logical to conflate these two issues, and ask:“Do patients with satisfactory weaning predictors usually avoid reintubation?” The only way to address this question in a scientific manner is to measure weaning predictors and extubate the patient without an intervening weaning trial. Zeggwagh et al[33] are the only group of investigators to undertake such a study.

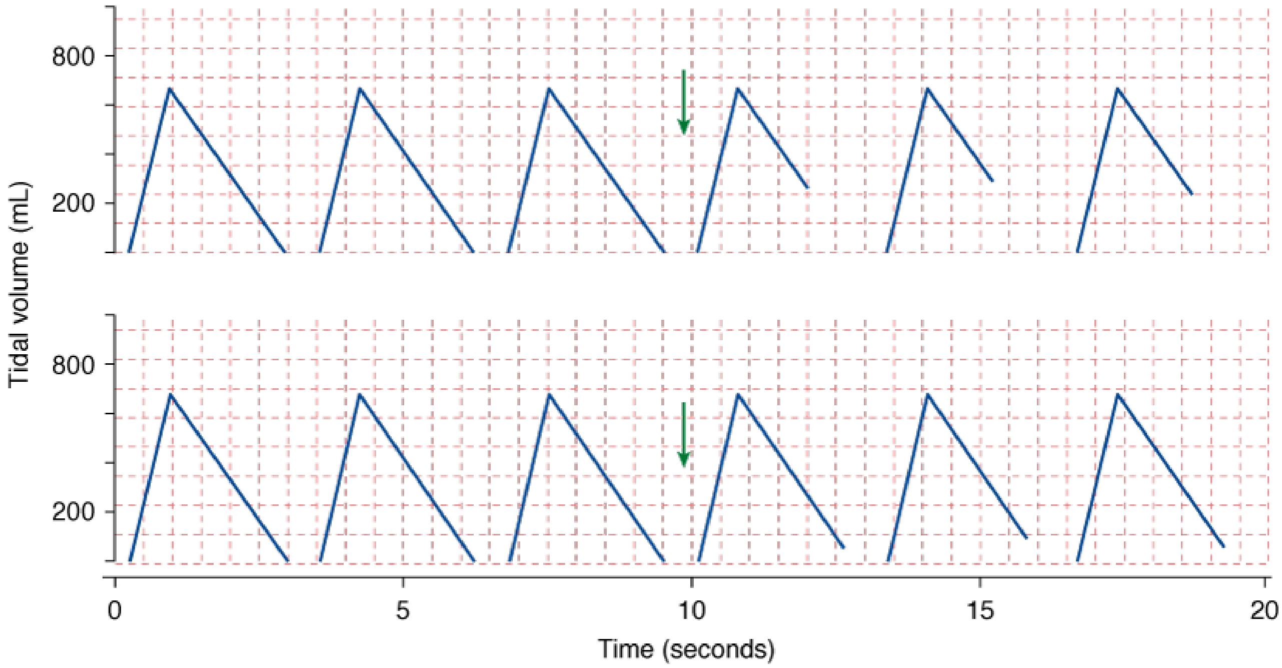
The investigators prospectively studied 101 patients (ventilated for 10.4 ± 10.3 days) at the point that their ICU physicians contemplated weaning. They measured a series of physiologic measurements during 2 minutes of spontaneous breathing; the results of these measurements were not communicated to the primary team. The team then extubated the patients without first undertaking any form of weaning trial. The extubation decision was made by the ICU team, based on the following criteria: improvement or resolution of the condition precipitating the need for mechanical ventilation; good level of consciousness with cessation of all sedative agents; temperature less than 38°C (100.4°F); respiratory frequency less than 35 breaths/min; S_{O_2} greater than 90% on $F_{I_{O_2}}$ equal to or less than 0.40; hemodynamic stability; and the absence of electrolyte disorders, acid–base disturbance, or hemoglobin less than 10 g/dL. Reintubation was necessary in 37% of the patients. Several variables predicted the need for reintubation with a reasonable degree of accuracy. For example, f/V_T had a sensitivity of 0.77 and a specificity of 0.79, with an area under a receiver operating curve (ROC) curve of 0.81 ± 0.06; maximum expiratory pressure had a sensitivity of 0.52 and a specificity of 0.92, with an area under an ROC curve of 0.73 ± 0.07. The investigators developed a model based on three variables: f/V_T, maximum expiratory pressure, and vital capacity. The area under the ROC curve for the model was 0.91 ± 0.04 for a development data series and 0.86 ± 0.06 for a validation data series.

This study by Zeggwagh et al[33] suggests that undertaking a weaning trial before extubation is useful because their rate of reintubation is about double that reported in studies in which weaning trials precede extubation. The clinicians in the study by Zeggwagh et al[33] based their decision for extubation on only the most rudimentary clinical assessment. If the clinicians had made the decision to extubate using more sophisticated physiologic predictors (but still did not undertake a weaning trial), it is likely that the rate of reintubation would have been lower. The accuracy of weaning predictors in this study contrasts sharply with their accuracy in studies in which the investigators permitted a weaning trial (which altered clinician’s extubation decisions) between measurement of the predictors and extubation (see Table 58-1 in Chapter 58).

Cuff-Leak Test

Some patients show satisfactory recovery of lung function but develop upper airway obstruction after extubation. The difficulty in defining the relationship between upper airway injury and postextubation stridor is that the presence of the endotracheal tube precludes direct visualization of the upper airway before extubation. Several investigators have tested the hypothesis that the volume of air leaking around the outside of an endotracheal tube on deflating the balloon cuff will be inversely related to the degree of laryngeal obstruction generated by laryngeal edema (Fig. 59-4). The idea was first reported by Adderley and Mullins[34] who studied thirty-one planned extubations in twenty-eight children with croup. After extubation, reintubation was required in 13% (3/23) of children who had an audible leak (on coughing or when plateau pressure was 40 cm H_2O), and reintubation was required in 38% (3/8) of children without a leak. The cuff-leak test has since been evaluated with varying degrees of scientific rigor (Table 59-3).

Figure 59-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Cuff-leak test. Schematic of tidal volume during inhalation (*up going*) and exhalation (*down going*) in two patients before and after deflation of the cuff on the endotracheal tube (*arrow). Upper panel:* The patient develops a large leak after cuff deflation (positive-test result), as signified by the expired tidal volume being markedly smaller than the inspired tidal volume. *Lower panel:* The patient develops a small leak after cuff deflation (negative-test result), as signified by the expired tidal volume being only approximately 40 to 75 mL or 6% to 12% less than the inspired tidal volume.

Table 59-3: Accuracy of the Cuff-Leak Test

Authors	No[a]	Leak Criterion	Outcome Criterion	Sensitivity	Specificity	PPV	NPV
Adderley et al[34]	31	Audible	Reintubation	0.80	0.50	0.87	0.38
Miller et al[35]	100	<110 mL	Stridor	0.67	0.99	0.80	NR
Engoren[90]	531	<110 mL	Leak <110 mL	0.00	0.96	0.00	1.00
Sandhu et al[19]	110	<10% inspV_T	Stridor or reintubation	NR	0.96	NR	NR
De Bast et al[22]	76	<15.5% inspV_T	Reintubation secondary to laryngeal edema	0.75	0.72	0.25	0.96
Jaber et al[20]	112	<130 mL or <12% inspV_T	Stridor and leak <130 mL or <12% of inspV_T	0.85	0.95	0.69	0.98
Maury et al[44]	115	Audible	Stridor	1.0	0.80	0.15	1.0
Kriner et al[91]	462	<110 mL	Stridor or reintubation	0.50	0.84	0.12	0.97
Chung et al[36]	95	<140 mL	Laryngeal edema on endoscopy	0.89	0.90	0.84	0.93

Abbreviations: insp, inspiratory; *NPV,* negative-predictive value; *NR,* not reported; *PPV,* positive-predictive value; *V_T,* tidal volume.

[a]Number of extubations.

Miller and Cole[35] undertook the first systematic evaluation of the cuff-leak test. They studied 100 intubations in eighty-eight ventilated patients. During assist-control ventilation, they noted that the set inspired tidal volume (V_T) and displayed expired V_T were always within 20 mL of each other (they did not state the volume setting). To measure the leak, they deflated the cuff and recorded expiratory V_T over the subsequent six cycles; they used the average of the three lowest values to calculate the leak. After extubation, 17% of patients required reintubation. Of the six patients who developed stridor, reintubation was required in three. The leak (measured during the 24 hours before extubation) was smaller in patients who subsequently developed postextubation stridor than in patients who did not: 180 ± 157 versus 360 ± 157 mL. ROC curve analysis indicated that a leak of less than 110 mL provided the best threshold for predicting postextubation stridor. A leak of less than 110 mL had a sensitivity of 0.67, a specificity of 0.99, and a positive-predictive value 0.80 (negative-predictive value was not reported).

A representative example of the methodology employed in subsequent investigations is illustrated by the approach used by Jaber et al[20] who evaluated the cuff-leak test in 112 patients. The leak was smaller in thirteen patients who developed postextubation stridor than in the ninety-nine patients who did not develop stridor after extubation: 59 ± 92 versus 372 ± 170 mL (with V_T 10 to 12 mL/kg during assist-control ventilation); expressed in terms of relative volumes, 9 ± 3% versus 56 ± 20%. Reintubation was required in 69.2% (9/13) of the patients with stridor as compared with 2.0% (2/99) of patients without stridor. They used ROC curve analysis to find the best threshold; they considered a true-positive test as a leak less than 130 mL or less than 12% of inspired V_T together with postextubation stridor. At this threshold, sensitivity was 0.85, specificity 0.95, positive-predictive value 0.69, and negative-predictive value 0.98.

A more rigorous methodology was employed by De Bast et al[22] who prospectively studied seventy-six patients who had been intubated 12 or more hours. They measured cuff leak as a percentage:

[(Expired V_T with cuff inflated − Expired V_T with cuff deflated)/Expired V_T with cuff inflated] × 100.

They took the average of six measurements that varied by less than 30%. The measurements were not available to the staff in charge of patients. If patients developed stridor associated with signs of respiratory distress within 24 hours of extubation and required reintubation, laryngeal edema was confirmed or excluded by fiber-optic examination before the reintubation, or by direct examination of the glottis during reintubation. The investigators excluded patients who were reintubated for reasons other than laryngeal edema.

Within the first 24 hours, 10.5% (8/76) of the patients of De Bast et al[22] required reintubation for laryngeal edema. These patients had smaller leaks than did the other patients: 9% (3, 18; twenty-fifth, seventy-fifth percentiles) versus 35% (13, 53; twenty-fifth, seventy-fifth percentiles). (Of the ten patients who developed postextubation stridor, eight were reintubated.) ROC curve analysis revealed that the best threshold was a leak of 15.5%. This threshold had a sensitivity 0.75, specificity 0.72, positive-predictive value 0.25, and negative-predictive value 0.96. The low positive-predictive value, 0.25, indicates that 75% of the patients with a leak less than 15.5% were successfully extubated (without laryngeal edema or requiring reintubation). Thus, a low leak volume should not be used to postpone extubation indefinitely. The negative-predictive value of 0.96 means that when patients exhibit a large leak (a negative test result), they are not likely to require reintubation (because of laryngeal edema). A leak greater than 23% excluded all patients who required reintubation because of laryngeal edema.

Chung et al[36] also used endoscopic examination of the upper airway in a study involving ninety-five patients. The average duration of translaryngeal intubation was quite long, 28.1 ± 17.6 (standard deviation [SD]) days. Employing video bronchoscopy, the Taiwanese investigators observed a high rate of severe laryngeal edema, 36.8% (35/95). They used ROC curve analysis to select the most discriminatory cuff-leak threshold. At a threshold volume of 140 mL, 39% (37/95) of patients had positive test results. The performance of the cuff-leak test was superior to that observed by most other investigators (see Table 59-3).

Prinianakis et al[37] undertook a study to determine the physiologic determinants of the volume being leaked when the cuff-leak test is performed. The volume of the leak is calculated by obtaining several measurements of expired V_T after deflation of the cuff and subtracting the average value from the set inspired V_T. With the cuff deflated, however, the inspiratory V_T reaching the alveoli is also decreased. Thus, the measured leak has both inspiratory and expiratory components. By adding an inspiratory pause, Prinianakis et al[37] measured expiratory leak unaffected by the inspiratory leak. In fifteen critically ill patients receiving neuromuscular blocking agents, the expiratory leak was consistently lower than the total leak. To elucidate the physiologic determinants of the expiratory leak, the investigators employed a mechanical lung model. The cross-sectional area around an endotracheal tube was not the only factor affecting the magnitude of a leak, it was also influenced by the inspiratory component. The total leak was inversely related to inspiratory flow and system compliance. Although this study explains some of the imprecision of the cuff-leak test, it does not mean that clinicians should start using a more pure measure of expiratory leak in everyday practice. Prinianakis et al[37] did not compare the accuracy of the more pure measure of expiratory leak in predicting the occurrence of postextubation laryngeal edema or the need for reintubation.

In summary, the studies of the cuff-leak test are of varying quality. The method for performing the test has not been standardized. In particular, none of the investigators addressed the setting of inspired V_T, which may influence the size of the leak. The method for quantifying the leak varies between absolute units (milliliters) and percentage of inspired V_T. The outcome criterion is not always clearly stated: rate of reintubation for any reason, occurrence of stridor of any severity, or occurrence of stridor that requires reintubation. The rates of stridor vary considerably among studies, suggesting that investigators used different criteria (admittedly, it is not obvious that severity of stridor can be graded in any reproducible manner). Only De Bast et al[22] and Chung et al[36] used an objective method (fiber-optic endoscopy) to verify the presence of laryngeal edema.

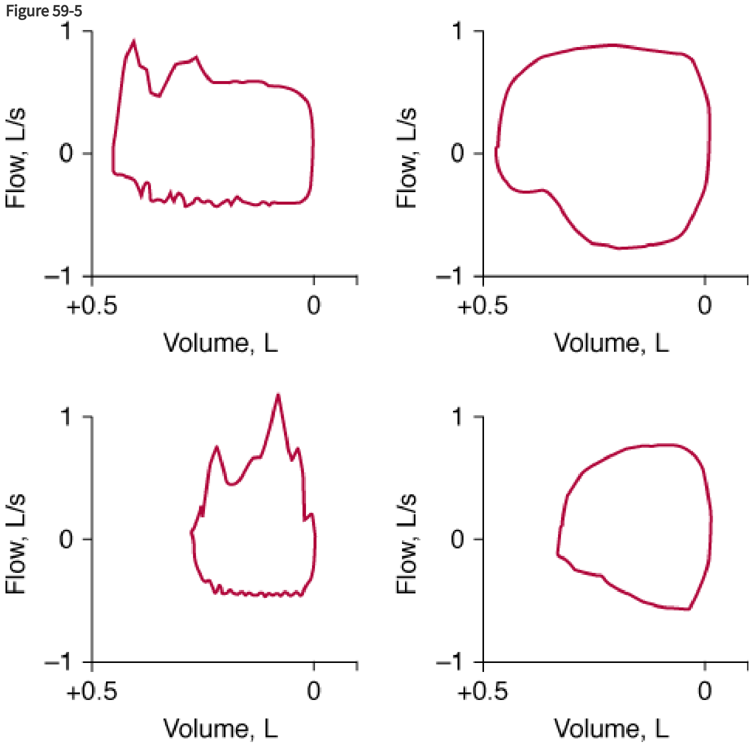
In some studies, it is not clear whether the investigators carefully excluded reasons for reintubation other than stridor. If a patient is reintubated because of left-ventricular failure, it is not logical to expect the cuff-leak test to predict such an event. What is the best reference standard? Should a true-positive test result be restricted to postextubation stridor that requires reintubation? Is it important to predict the development of postextubation stridor that will respond to aerosolized epinephrine? The manner of reporting test performance is not consistent. The thresholds for defining a significant leak vary. All calculations of test performance are inevitably overestimates, because none of the investigators split their data set into training and validation subsets. A fundamental problem with all of the studies is that the leak criterion used to define a positive test has been based on the assumption that the clinical consequences of false-positive and false-negative results are equal. Instead, physicians need to customize the threshold value according to the clinical circumstances. In cases where tracheal intubation is expected to be difficult, physicians may use a higher leak volume as the threshold for a positive test result. Conversely, when there is greater fear of the hazards of delayed extubation, physicians may accept a lower leak volume as the threshold for a positive test result.

Secretions and Cough

A proportion of patients fail either a weaning attempt or an extubation attempt because of excessive airway secretions. This proportion varies among reports, largely because there is no consistent definition of “excessive secretions” or even how best to quantify secretions. If one quantifies secretions according to the volume obtained by suctioning over a fixed time interval, a patient who coughs and expels secretions without difficulty may get classified as having a greater secretion problem than a patient who has thick viscid secretions that cannot be dislodged from the lower airways.

The act of suctioning per se is associated with a number of serious complications, such as life-threatening hypoxemia, mucosal trauma, hemorrhage, bronchoconstriction, atelectasis, cardiac arrhythmias, and even cardiac arrest.[38–41] Jubran and Tobin[42] investigated the possibility that a sawtooth pattern on the flow-volume curve might provide a noninvasive means of detecting secretions. In fifty intubated patients, the presence of a sawtooth pattern on flow-volume curves recorded during 1 minute of spontaneous breathing was a strong predictor of the presence of secretions (positive-predictive value: 94%), and the absence of this pattern suggested that secretions were unlikely to be present (negative-predictive value: 77%) (Fig. 59-5). Expressed in terms of likelihood ratios, a sawtooth pattern was approximately six to eight times more likely to be found in patients who had secretions than in patients without secretions. Conversely, a smooth flow-volume curve was about one-quarter as likely to be found in patients with secretions as in those without secretions. Clinical examination had much higher false-positive and false-negative rates (42% and 43%, respectively) than the flow-volume curves (12% and 14%, respectively). Accordingly, reliance on clinical examination will lead to unnecessary suctioning in patients without secretions and inadequate suctioning in patients with secretions.

Figure 59-5



Left panels, upper and lower: Inspiratory and expiratory flow-volume curves obtained in two patients with secretions exhibit a sawtooth pattern. *Right panels, upper and lower*: Flow-volume curves obtained in two patients without secretions have a smooth contour. (Modified, with permission, from Jubran and Tobin.[42])

Measurement of secretions has been evaluated as a predictor of postextubation distress. Khamiees et al[43] attempted to quantify cough strength by placing a white card at 1 to 2 cm from the end of the endotracheal tube and requesting the patient to cough as many as three to four times just before extubation. Any wetness on the card was classified as a positive test (assessment was made by a single observer). This test was seen as a test of cough strength and not of the amount of secretions present. They studied 100 extubations in ninety-one patients;

eighteen patients were classified as extubation failures and eleven were reintubated within 72 hours of extubation (the criteria for classifying the other seven patients as extubation failures are not clear). Extubation failure was three times more likely in patients with a negative white-card test (no secretions coughed onto the card). Three other measures also predicted extubation failure. Extubation failure was four times more likely among patients who had a weak or absent cough than in patients with a moderate or strong cough. Extubation failure was eight times more likely in patients classified as having moderate or abundant secretions by the nursing staff in the 4 to 6 hours preceding extubation than in patients with absent or mild secretions. Extubation failure was sixteen times more likely among patients whose secretions required suctioning every 2 hours or less.

Smina et al[16] of the same research group studied the ability to predict reintubation after 115 extubations in ninety-five patients (on this occasion, the investigators specify that extubation failure was synonymous with reintubation). Reintubation within 72 hours was required in 11.3%. The magnitude of secretions was not associated with extubation failure, which contrasts with their previous report that extubation failure was increased sixteen times if patients required frequent suctioning. This failure to reproduce an earlier observation illustrates a common problem with all studies of clinical predictors. After the report of findings of an initial study, physicians modify their approach to a particular problem; in statistical terminology, clinicians change their pretest probability (based on the report). When a research team reinvestigates the same phenomenon in a new group of patients, they fail to reproduce the original finding. Many patients in the second study who had secretions were not advanced to extubation, leading to a decrease in the number of true-positive test results. The study of Khamiees et al[43] indicated that frequent suctioning was associated with a high reintubation rate. Smina et al,[16] however, found that only 4% of patients in their follow-up study had greater than 20 mL of secretions per hour before extubation. Clearly, physicians in the second study had reduced the number of extubations attempted in patients with larger volumes of secretions. The physicians had altered their pretest probability of extubation failure based on the need for frequent suctioning. The physicians refused to advance such patients to extubation (test-referral bias), and thus, the results of the study give an erroneous impression that frequent suctioning is not a good predictor of reintubation.

Maury et al[44] hypothesized that the occurrence of a cough in the first few seconds after complete deflation of the cuff on an endotracheal tube results from secretions, which had previously accumulated above the cuff and had now passed into the lower airways. They reasoned that the presence of a cough associated with respiratory gurgling (heard without a stethoscope and related to secretions) upon cuff deflation would indicate patency of the upper airway, whereas the absence of a cough would suggest upper airway edema—the same line of thinking on which the cuff-leak test is premised. They tested the hypothesis during 115 extubations in ninety-nine patients. The absence of a cough following cuff deflation was a strong predictor of postextubation stridor (negative predictive value: 0.98); positive-predictive value, however, was only 0.12.

Neurologic Assessment

Some ventilated patients demonstrate good respiratory function and tolerate a T-tube trial without distress, yet their physicians are reluctant to extubate them because they fear that the patients will not be able to protect their airway after extubation. Although widely used, the term *protecting the airway* is rarely clearly defined. We take it to mean that a patient has unsatisfactory neural control over the airway, such that the tongue (of a recumbent patient) may fall back and occlude the airway lumen (as happens in patients with sleep apnea), or the patient has impaired laryngeal (and other upper airway) reflexes, placing the patient at risk of aspiration of secretions (from the mouth and airways) or ingested food.

Concern about protecting the airway most often arises in a patient with evidence of brain injury. Three groups of investigators have studied the role of brain function in patients being considered for extubation. The most rigorous study is that by Coplin et al,[45] who studied 136 brain-injury patients. They evaluated patients for extubation using a broad screen: absence of neurologic deterioration on physical examination; intracranial pressure less than 20 mm Hg; satisfactory oxygenation, lung mechanics, f/V_T, blood pressure, and heart rate; and absence of a specific indication for mechanical ventilation (such as surgery planned within the subsequent 72 hours).

Of the 136 patients, 72.8% (99/136) were extubated within 48 hours of meeting the above readiness criteria; the other 27.2% (37/136) remained intubated for a median of 3 days (range: 2 to 19 days). They defined a delay in extubation as the time from meeting readiness criteria to the time of extubation, but subtracted 48 hours (to allow for time needed for communication among caregivers). Neurologic evaluation based on the Glasgow Coma Scale (GCS) score was performed daily. (This scale ranges from 3 to 15, with 15 indicating the best brain function and 3 indicating severe brain dysfunction; a score ≥13 indicates *possible* mild brain injury, 9 to 12 moderate injury, and ≤8 severe brain injury.)

Sixty patients were judged comatose (score ≤8 on the GCS) on the day of meeting extubation readiness criteria; extubation was delayed in 48.3% (29/60) as compared with 10.5% (8/76) of patients not classified as comatose. The two groups, however, exhibited considerable overlap. Of sixty comatose patients (score ≤8), 51.7% (31/60) were extubated without delay. Indeed, among patients with more severe brain injury, with GCS scores of 4 and lower, 40% (4/10) were extubated without delay. Conversely, 10.5% (8/76) of patients with scores ≥9 experienced delayed extubation. Coplin et al[45] assessed whether the decision to extubate was influenced by change in neurologic status over time: 57% (21/37) of patients improved between the day of meeting the readiness criteria and extubation, but the remaining 43% showed no change or deterioration in neurologic function.

Absence of a gag reflex has been considered a contraindication to extubation in the past. Approximately 20% of healthy people, however, do not have a gag reflex, and aspiration pneumonia may still occur in people who do.[46] The importance of the gag reflex in the extubation period was addressed by Coplin et al:[45] 89% (32/36) of patients with a weak or absent gag reflex were successfully extubated.

A major reason to postpone extubation in a neurologically impaired patient is the fear of aspiration pneumonia. The occurrence of pneumonia, however, was higher in patients experiencing delayed extubation: 38% versus 21%.[45] These patients also had longer stays in the ICU (8.6 vs. 3.8 days) and in hospital (19.9 vs. 13.2 days). Based on their data, Coplin et al[45] concluded that a depressed level of consciousness should never be used as the sole indication for prolonged intubation.

In contrast to Coplin et al,[45] Namen et al[13] concluded that the GCS helps in predicting successful extubation. They studied 100 brain-injury patients, whose mechanical ventilations were managed by neurosurgeons. On ROC curve analysis, a score of ≥8 on the GCS provided the best discrimination of extubation outcome. Extubation was successful in 36% of patients with GCS scores ≤7 as compared with 75% of patients with a score ≥8. The area under the ROC curve for GCS score, however, was only 0.681. A more fundamental problem is that twenty-two patients were extubated as part of the withdrawal of life-support therapy (all these patients died). Because these patients were not reintubated, it appears that the authors classified them as extubation successes. Irrespective of how these twenty-two patients were classified, it is impossible to interpret data on extubation predictors where half of the extubations arose from a decision to withdraw life support.

The studies of Coplin et al and Namen et al were conducted in patients with brain injury, whereas Salam et al[47] studied neurologic function as a predictor of reintubation in eighty-eight medical-cardiac ICU patients who underwent 100 extubations. Neurologic performance was quantified by requesting patients to perform four simple tasks[48]: to open their eyes, to follow an observer with their eyes, to grasp the observer's hand, and to stick out their tongue. Reintubation within 72 hours was required in 15.9% (14/88) of the patients. Patients tolerating extubation performed a higher number of tasks than did the reintubated patients: 3.8 ± 0.1 versus 2.9 ± 0.5. Patients who were unable to complete all four tasks were 4.3 times more likely to require reintubation than were patients who could complete all four tasks. The failure to perform any of the four tasks had a sensitivity of 0.42 and specificity of 0.91 in predicting reintubation.

Treatment of Postextubation Laryngeal Edema

Epinephrine

Trials of therapies for postextubation respiratory distress have largely been confined to patients with laryngeal edema. For decades, patients with postextubation stridor and other upper airway disorders have been treated with aerosolized racemic epinephrine. Racemic epinephrine consists of equal amounts of the dextro (d)-isomer and levo (l)-isomer. Most of epinephrine's pharmacologic action results from the levo-isomer, which is thirty times more potent than the dextro-isomer.[49] Popularity of the more expensive racemic form is based on the supposition that it produces epinephrine's vasoconstrictor action without rebound vasodilation; thus, less tachycardia, hypertension, and tremor is expected with aerosolized racemic epinephrine than with levo-epinephrine. The stated different actions, however, may have arisen from comparisons of inappropriate dosages.

Nutman et al[50] randomized twenty-eight children with postextubation stridor (average age: 1 year) to receive 0.25 mL of either 2.25% racemic epinephrine or 1% levo-epinephrine, each diluted with 2 mL of isotonic saline. These dosages were selected to reflect the relative potency of the two compounds. Each was delivered over 15 minutes by face mask. Both groups exhibited significant improvement: 71.4% of the levo-epinephrine group and 76.9% of the racemic group exhibited decreases of 2 points on an 8-point stridor score after 40 minutes. By 8 hours, the stridor score was less than 1 in both groups. These data reveal that levo-epinephrine is as effective as the more expensive racemic epinephrine in children with postextubation stridor, although it is not known how likely the stridor would have resolved without any aerosolized therapy.

Glucocorticoids

The ability of dexamethasone to prevent postextubation stridor in children has been evaluated by three groups of investigators. In the first two studies, dexamethasone (0.5 mg/kg every 6 hours for 6 doses, beginning 6 to 12 hours before extubation) was compared against placebo. In their study of 153 children, ranging from under 1 year old to older than 5 years, Tellez et al[51] observed postextubation stridor requiring therapy in 21.1% (16/76) of the dexamethasone group and 29.9% (23/77) of the control group. Reintubation was required in 11.8% (9/76) of the dexamethasone group and 5.2% (4/77) of the control group. In a study of sixty-six children younger than 5 years old, Anene et al52 observed stridor at 10 minutes in 45.2% of the dexamethasone group and 87.5% of the control group. Epinephrine aerosol was required in 12.9% (4/31) of the dexamethasone group and 68.8% (22/32) of the control group. Reintubation was required in 0/31 of the dexamethasone group and 21.9% (7/32) of the control group. Apart from the different conclusions on the benefit of dexamethasone, the different outcomes in the two control groups is striking: postextubation stridor requiring therapy was reported in 68.8% of the patients of Anene et al[52] as contrasted with 29.9% of the patients of Tellez et al;[51] the respective rates for reintubation were 21.9% and 5.2%.

A more recent trial was undertaken by Cesar and de Carvalho[53] who randomized sixty-four children, ranging in age from younger than 1 year to 12 years, to intravenous dexamethasone (0.2 mg/kg every 6 hours), with or without nebulized levo-epinephrine (0.5 mg/kg every 4 hours), versus nebulized or intravenous isotonic saline solution in the control groups. Neither dexamethasone nor nebulized levo-epinephrine, alone or in combination, decreased the frequency of laryngeal edema.

Early randomized controlled trials of the administration of glucocorticoids before extubation in adult patients were negative. Darmon et al[24] randomized 663 patients to a bolus of dexamethasone 8 mg or placebo 1 hour before extubation. The overall incidence of laryngeal edema was 4.2% (28/663), and reintubation was necessary in 1% (7/663). No difference was seen between patients receiving dexamethasone or placebo. Ho et al[23] randomized seventy-seven intubated patients to hydrocortisone 100 mg or placebo, given 1 hour before extubation. Overall, 22% (17/77) developed stridor within 24 hours, and only one patient required reintubation. Outcomes were equivalent for patients receiving hydrocortisone and placebo. More recent trials indicate that glucocorticoids can be beneficial, especially when administered for 12 or more hours before extubation to patients at high risk of postextubation laryngeal edema.

Cheng et al[54] selected a group of high-risk adults intubated for more than 24 hours with a decreased cuff-leak volume (<24% of inspired V_T). Patients were randomly assigned to multiple doses of methylprednisolone (40 mg every 6 hours for 4 doses) before a planned extubation, a single dose of methylprednisolone 24 hours before the planned extubation, or placebo (4 injections of saline every 6 hours). Postextubation stridor developed in 30.2% of placebo group, and 18.6% required reintubation. Treatment with either a single dose or multiple doses of methylprednisolone decreased the rate of postextubation stridor, 11.6% and 7.1%, respectively, and also the rate of reintubation, 4.7% and 7.1%, respectively. There was no difference between the effects of single and multiple doses of methylprednisolone.

In 761 adult patients admitted to fifteen medical-surgical ICUs and receiving more than 36 hours of mechanical ventilation, François et al[21] conducted a double-blind controlled trial. Patients were randomized to intravenous methylprednisolone—initiated 12 hours before planned extubation at a dose of 20 mg and continued every 4 hours with the last injection immediately before tube removal (total dose: 80 mg)—or placebo. Methylprednisolone decreased the incidence of postextubation laryngeal edema (11/355 [3%]vs. 76/343[22%]) and the overall reintubation rate (13/355 [4%]vs. 26/343 [8%]). The need for reintubation as a result of laryngeal edema was all but eliminated (1/355 [0.3%]vs. 14/343 [4.1%]). The number needed to treat to prevent one reintubation caused by laryngeal edema was slightly more than twenty-six. Although the number needed to treat was large, the number of adverse events in the intervention group was small; thus, methylprednisolone appears have a satisfactory risk-to-benefit ratio in this setting.

Lee at al[55] conducted a randomized, double-blind trial in eighty-six patients who were at high risk of postextubation airway obstruction, in that they had a cuff-leak volume of less than 110 mL. Patients were randomly assigned to receive placebo or dexamethasone 5 mg every 6 hours for 24 hours, and were extubated 24 hours later. Compared with placebo, dexamethasone produced an increase in cuff-leak volume, approximately 115 mL at 24 hours versus approximately 40 mL, and decreased the incidence of postextubation stridor, 10% versus 27.5%. The rate of reintubation did not differ between the dexamethasone and placebo group: 2.5% versus 5%.

Cheng et al[56] undertook a second randomized, controlled trial, and like their previous trial they selected a group of high-risk adults with a decreased cuff-leak volume (<24% of inspired V_T). Four hours before planned extubation, thirty-eight patients received a single injection of methylprednisolone, 40 mg, and thirty-three patients received normal saline. Methylprednisolone resulted in a lower incidence of postextubation stridor (15.8% vs. 39.4%) and lower reintubation rate (7.9% vs. 30.3%). The beneficial effects were accompanied by the upregulation of interleukin (IL)-4 and IL-10 and the downregulation of IL-6 and IL-8.

Jaber et al[57] conducted a meta-analysis of randomized controlled trials of the effects of glucocorticoids on postextubation stridor and reintubation. Fifty-six trials have been conducted, and Jaber et al[57] selected seven that satisfied predefined methodologic criteria. These included the above discussed trials of Darmon et al,[24] Ho et al,[23] Cheng et al,[54] François et al,[21] and Lee et al,[55] and two other studies—Shih et al[58] and Cheng et al[59]—that had only been presented in abstract form; the abstracted Cheng et al study has since been published as a full report.[56] Among an aggregate of 1846 patients, those receiving glucocorticoids had a lower incidence of stridor (relative risk [RR]: 0.48;95% confidence interval [CI] 0.26 to 0.87) and lower rate of reintubation (RR: 0.58; 95% CI 0.41 to 0.81). Subgroup analysis revealed that glucocorticoids decreased the risk of reintubation more strikingly in high-risk patients, defined as those with an abnormally low volume on the cuff-leak test. In this subgroup, the rate of reintubation decreased from 19.8% to 8.6% (RR: 0.38; 95% CI 0.21 to 0.72). The benefit of glucocorticoids was not as striking in patients who were not selected on the basis of risk of reintubation (RR: 0.67; 95% CI 0.45 to 1.00).

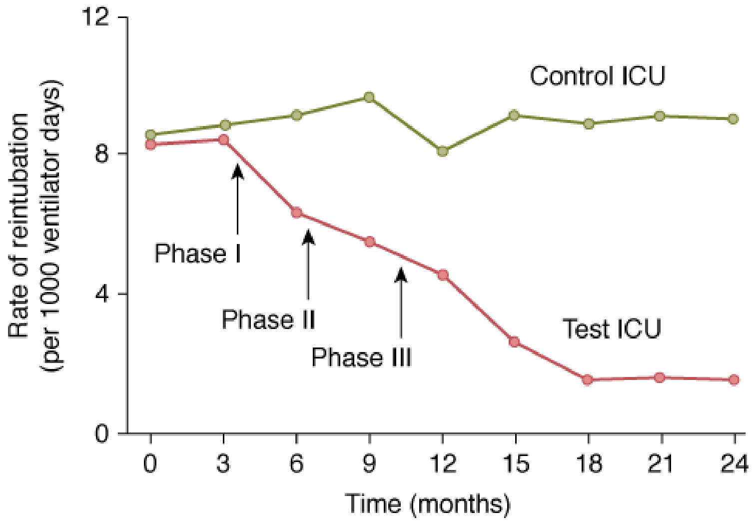
In patients at risk of postextubation laryngeal edema, glucocorticoids are presumed to exert their beneficial effect by inhibiting the release of inflammatory mediators and decreasing capillary permeability.[57] The initial antiinflammatory effects commence 1 to 2 hours after intravenous administration and are maximal between 2 and 24 hours later. Consistent with this knowledge, clinical trials in which glucocorticoids were administered 1 hour or less before planned extubation did not decrease postextubation stridor or the rate of reintubation benefit,[23,24] whereas benefit was observed in trials when glucocorticoids were administered as long as 24 hours before extubation.[21,54,55] It has been proposed that physicians should select patients for glucocorticoid therapy on the basis of the cuff-leak test and then wait for 24 hours to allow the agent to take effect. This proposal, however, is problematic, as it would unnecessarily prolong intubation by 24 hours in three of every four patients with a low cuff-leak volume. If the recent observation of Cheng et al[56] is confirmed—that a single dose of methylprednisolone, 40 mg, administered 4 hours before extubation is beneficial in patients with a low cuff-leak volume—such an approach would represent a satisfactory risk-to-benefit ratio.

Intervention to Reduce Need for Reintubation

Pronovost et al[60] investigated the effect of a new quality improvement intervention on the frequency of reintubation in their surgical ICU. Their study was undertaken in three phases. In the first phase, they identified risk factors associated with reintubation. In the second phase, ten ICU physicians engaged in a Delphi process, attempting to develop clinical practice guidelines. The physicians could not agree on priorities; they concluded that there were too many conditional probabilities that would require the use of so many decisional nodes, as to render a guideline impractical. This group used data collected in the first phase to develop a preextubation worksheet designed to highlight factors associated with reintubation. In discussion with a patient's nurse and respiratory therapist, physicians listed several variables on this form. Variables included whether intubation had been difficult, ventilator settings, oxygen saturation, arterial blood-gas results, f/V_T, frequency of suctioning, and mental status. In the third phase of the study, they studied the effect of implementing the preextubation worksheet on physician behavior. Staff was instructed not to extubate a patient if the worksheet had not been filled out.

The rate of reintubation before the quality improvement intervention was 8 per 1000 ventilator days. After the intervention, reintubation decreased to 1.5 per 1000 ventilator days. In another (control) surgical ICU in the same hospital, the rate of reintubation did not change over the same period (Fig. 59-6). The average duration of mechanical ventilation decreased from 4.6 days in the first phase of the study to 3.4 days in the third phase. Three factors were associated with reintubation: suctioning more often than every 4 hours (OR: 11.3); being agitated or sedated versus alert (OR: 4.5); and S_{O_2} less than 95% on $F_{I_{O_2}}$ 0.40 (OR: 4.0). The accuracy of these and other variables is clearly influenced by test-referral bias. The authors speculate that the preextubation worksheet helped to bring directly to the staff's attention the most important factors to consider when deciding to extubate.

Figure 59-6



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

The rate of reintubation per 1000 ventilator days in a surgical ICU that introduced a new quality improvement intervention and in a control surgical ICU. The intervention was introduced in three phases in the study ICU. The intervention led to a decrease in the rate of reintubation from 8 to 1.5 per 1000 ventilator days. The rate of reintubation did not change in the control ICU. (Used, with permission, from Pronovost et al.[60])

Noninvasive Ventilation in Weaning and Extubation

Many investigators have sought to see if NIV can assist with the process of weaning and extubation. Some investigators have employed NIV as a bridge to ventilator discontinuation, where NIV is used in a patient who is believed to be not quite ready for extubation.[61] Other investigators have employed NIV as rescue therapy, where NIV is used in a patient who develops respiratory distress several hours after extubation.[6] The various studies that have addressed the use of NIV for these two purposes were discussed in considerable depth in the second edition of this book.

Three groups of investigators—Nava et al,[61] Girault et al,[62] Ferrer et al[63]—have evaluated the use of NIV as a weaning aid. These investigators instituted NIV at a point where a patient had failed a T-tube trial; patients typically received inspiratory positive airway pressure of 10 to 20 cm H_2O, and expiratory positive airway pressure of to 5 cm H_2O.[61,63] Patients in the intervention arm were placed on NIV immediately following extubation and were compared to a control group of patients who were reconnected to the ventilator and underwent conventional weaning (daily T-tube trials). Although the particularities of benefit differed among the three studies, all three research groups observed that the use of NIV as a bridge or weaning aid led to fewer days of mechanical ventilation, and two of the groups[61,63] observed a decrease in mortality as compared with conventional weaning.

These findings contrast with the use NIV as rescue therapy. Keenan et al[64] observed that institution of NIV at the point when a recently extubated patient developed respiratory distress, typically some hours after extubation, did not result in improved clinical outcome. Esteban et al[6] also employed NIV as rescue therapy in patients who developed postextubation distress. In contrast with all previous studies, they reported that NIV resulted in a greater mortality than observed with usual care: 25% versus 14%. Questions have been raised about the limited expertise of clinicians using NIV in the latter study and also that the employed level of pressure support was not reported. A striking feature was the outcome in twenty-eight patients in the usual care who crossed over to NIV. Despite being the sickest subgroup of the 107 patients in the usual-care group, mortality in the crossover subset was 11%—the lowest mortality of all groups requiring ventilator support.

Another feature that differs between the three studies with favorable outcomes[61–63] and the two studies with unfavorable outcomes[6,64] was that only 10% to 11% patients in the unfavorable studies had chronic obstructive pulmonary disease as contrasted with 44.2%,[63] 51.5%,[62] and 100%[61] in the studies with favorable outcomes. Patients who have hypercapnia as a result of chronic respiratory disorders, typically chronic obstructive pulmonary disease, have been shown repeatedly to exhibit an especially favorable response to NIV, and thus it becomes i an extubation aid in such patients.

In 106 patients with chronic respiratory disorders who had received mechanical ventilation for at least 48 hours and who exhibited a partial pressure of arterial carbon dioxide (Pa_{CO_2}) greater than 45 mm Hg during a T-piece weaning trial, Ferrer et al[65] randomly allocated fifty-four patients to NIV for the next 24 hours and fifty-two patients to conventional management. The primary end point was avoidance of respiratory failure (defined on meeting a number of predesignated laboratory or clinical criteria) within 72 hours following extubation. Such respiratory failure arose in fewer patients allocated to NIV than in those assigned to conventional management: eight of fifty-four (14.8%) versus twenty-five of fifty-two (48.1%); OR: 5.32 [95% CI 2.11 to 13.46], $p < 0.0001$).

When patients in either the NIV or control group met criteria for respiratory failure following extubation, but did not fulfill criteria for immediate reintubation, NIV was employed as rescue therapy. Among this subgroup, NIV resulted in avoidance of reintubation in two of seven patients assigned to NIV and fifteen of twenty controls. Overall, NIV was independently associated with a lower risk of respiratory failure after extubation (adjusted OR: 0.17 [95% CI 0.06 to 0.44]; $p < 0.0001$). Ninety-day mortality was lower in patients assigned to NIV than in those allocated to conventional treatment: six of fifty-four (11.1%) versus sixteen of fifty-two (30.8%); $p = 0.015$.

Given the impressive results achieved in this trial, it is important to note details of the way that Ferrer et al[65] employed NIV. The investigators used a ventilator specifically designed for NIV (BiPAP Vision, Respironics, Murrysville, PA), which provided effective compensation for leaks and real-time assessment of mask pressure. Inspiratory positive airway pressure was adjusted (12 to 20 cm H_2O) to achieve a respiratory rate less than 25 breaths/min (mean pressure: 17 ± 3 [SD] cm H_2O), expiratory positive airway pressure was fixed at 5 to 6 cm H_2O, and inspired oxygen concentration was titrated to achieve oxygen saturation of more than 92%. NIV was delivered for as much time as possible during the 24 hours that followed extubation (mean: 18 ± 7 [SD] hours); meals were not allowed during the first 24 hours.

In summary, data to date support the use of NIV as an aid to weaning patients who have chronic respiratory disorders and who exhibit hypercapnia during a weaning trial. In these patients, NIV appears to be beneficial as a bridge to ventilator discontinuation and also as rescue therapy in patients who develop respiratory failure following extubation. In patients who have underlying disease states other than a chronic respiratory disorder or who do not exhibit hypercapnia during a weaning trial, the role of NIV as a weaning aid is less clear. If it is to be used, it is likely to achieve greater success if instituted immediately after extubation[61–63] than employed as rescue therapy in patients who develop respiratory failure several hours after extubation.[6,64]

Unplanned Extubation

Chapter 39 discusses unplanned extubation in detail. The subject is of particular importance for research on weaning because of the insight it may offer into unnecessary delays in the discontinuation of ventilation. The reported frequency of unplanned extubation ranges from less than 1% to 42% of intubated patients; a frequency of 8% to 12% is most commonly reported (Table 59-4). Unplanned extubation is divided into two major subsets. Self-extubation refers to the deliberate removal of an endotracheal tube by a patient for whom the physician considers intubation beneficial. Accidental extubation refers to inadvertent extubation by a caregiver during a bedside procedure. Both of these situations differ from the usual planned extubation. Between 63% and 95% of unplanned extubations are self-extubations.

Table 59-4: Frequency of Unplanned Extubation and Subtypes

Author	Total Number[a]	Unplanned Extubation	Self-Extubation		Accidental Extubation		
		Percent of Total	Reintubation (%)	Percent of Unplanned	Reintubation (%)	Percent of Unplanned	Reintubation (%)
Zwillich et al[92]	354	8.5	47	30	47		
Stauffer et al[93]	226	12.8	NR				
Whelan et al[94]	319	7.2	78.2	91.3		8.7	
Tindol et al[95]	460	2.8	46.2	92.3		7.7	
Listello et al[96]	NR	NR	48.2				
Tominaga et al[72]		15.2	NR				
Boulain[97]	426	10.8	60.9				
Betbese et al[69]	750	7.3		78			
Chevron et al[98]	414	16	37.0	87			
Jiang et al[99]	97	42	37.8				
Kapadia et al[100]	5046	0.7					
Epstein et al[71]	682	11	56.0	94.7		5.3	
De Lassence et al[8]	750	8	76.7	63.3	63	36.7	100
Esteban et al[101]	5183	3.4	41.3				
Moons et al[67]	627	26	4.2	76.9	45	23.1	100
Thille et al[31]	340	9.1	64.5	67.7	47.6	32.2	100

Abbreviation: NR, not reported.

[a] Total number of ventilated or intubated patients.

The rate of reintubation after unplanned extubation ranges from 4% to 78%, with many investigators reporting rates of 35% to 60%. These rates are considerably higher than reintubation rates of 10% to 20% after planned extubation. The rate of reintubation after accidental extubation is especially high, 83%[66] to 100%.[8,67] That 40% or more of self-extubated patients do not require reintubation is often used as evidence that physicians delay extubation unnecessarily. Extubation is undoubtedly delayed in some patients, but the two groups of patients may not be strictly comparable. Patients who have enough strength—and wit—to self-extubate may be less sick than the average ventilated patient, and thus would be expected to experience a lower reintubation rate. Moreover, it is totally unclear as to what the source of the delay is in these self-extubated patients. Is it caused by clinicians who are not aware of a patient's readiness for extubation because they have yet to measure weaning predictor tests? Is it because the predictor test readings represent false-negative results? Is it because the physician interprets the predictor test results incorrectly? Is it because the physician delays in ordering a weaning trial?[68] Is it because the physician delays extubating a patient who has passed a weaning trial?[68] A considerable body of data[13,68] suggests that the most likely source of delay is in ordering a weaning trial and extubating a patient who has passed the trial.

The rate of reintubation after unplanned extubation is lower in patients who have already entered the phase of weaning than in patients who are still receiving full ventilator support. Betbese et al[69] reported that 16% of thirty-two patients who experienced unplanned extubation during weaning required reintubation, as contrasted with 82% of twenty-seven patients who experienced unplanned extubation while receiving full ventilator support. Razek et al[70] reported reintubation after unplanned extubation in 15.2% of thirty-three weaning patients and 60.7% of twenty-eight patients requiring full ventilator support. And Epstein et al[71] reported reintubation after unplanned extubation in 30% of thirty-three weaning patients and 76% of forty-two patients requiring full ventilator support. Epstein et al[71] noted that patients who tolerated unplanned extubation tended to have shorter time from the onset of weaning to extubation than did a control group, 0.9 versus 2.0 days ($p = 0.06$), although the overall duration of mechanical ventilation did not differ.

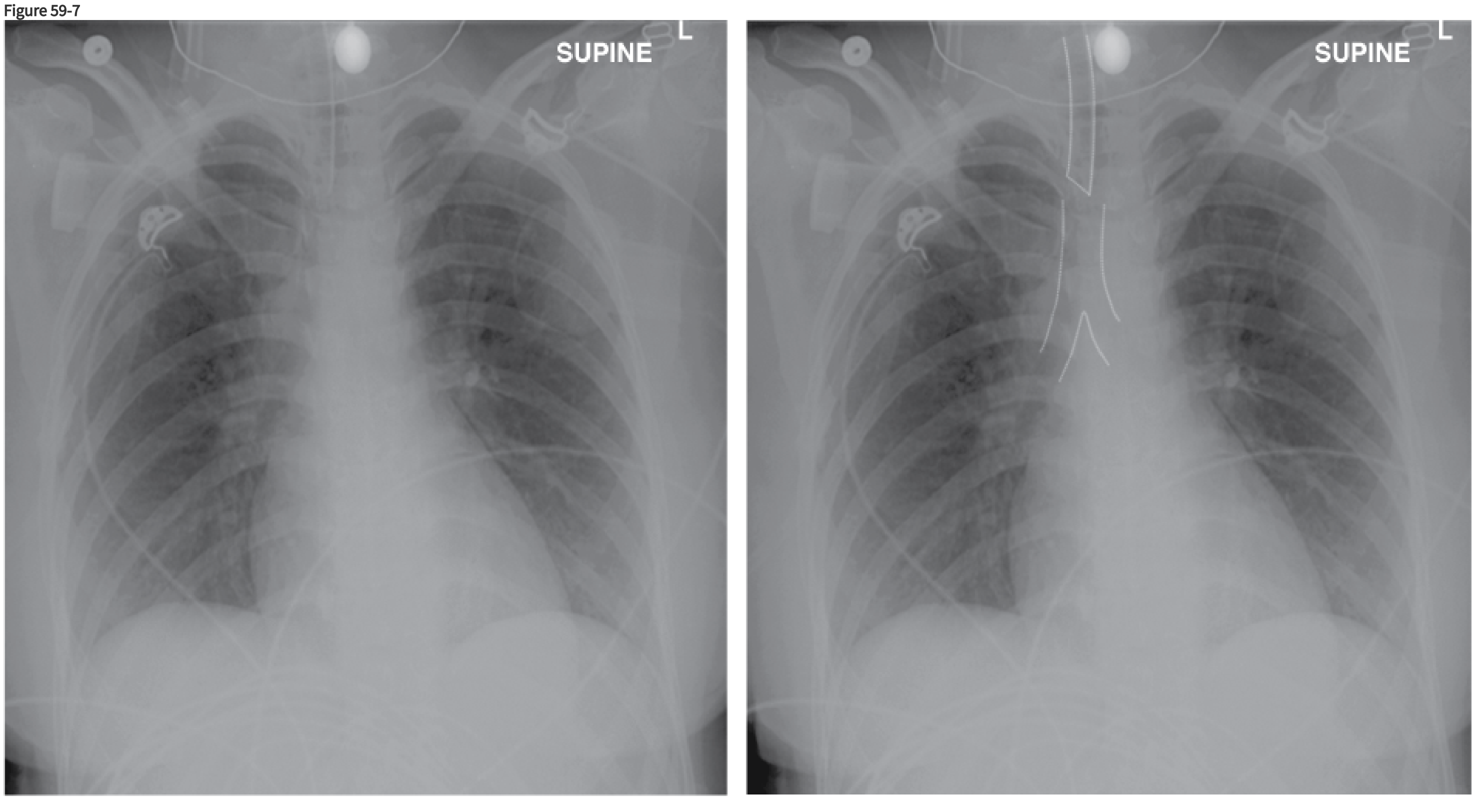
Moons et al[67] found that most unplanned extubations occurred on the same day that extubation had been planned (42.9%), or the subsequent (42.9%) or preceding day (14.3%). Compared to a control group of forty-eight patients that did not experience unplanned extubation, a study group of twenty-six patients experiencing unplanned extubation had a lower Ramsay Sedation Scale score, 2 (1, 5; twenty-fifth, seventy-fifth percentiles) versus 5 (3.3, 6; twenty-fifth, seventy-fifth percentiles), and a higher GCS score, 12 (8, 13; twenty-fifth, seventy-fifth percentiles) versus 4 (3, 7; twenty-fifth, seventy-fifth percentiles). The difference between the two groups was confined to the subgroup experiencing deliberate self-extubation; patients experiencing accidental extubation were comparable to the control group. In a multiple logistic regression analysis, deliberate self-extubation was associated with a lower sedation score (OR: 2.04) and higher level of consciousness on the GCS (OR: 1.40). The model explained 67.3% of variance of deliberate self-extubation. These data suggest that the weaning period is a high-risk time for deliberate self-extubation because patients are receiving less sedative agents and have a higher level of consciousness.

De Lassence et al[8] undertook a prospective multicenter study of the morbidity and mortality associated with unplanned extubation. Compared with forty-five of 690 patients who required reintubation after planned extubation, the forty-six (of sixty) patients who were reintubated after an unplanned extubation experienced a longer duration of mechanical ventilation, 17 versus 6 days; longer ICU stay, 22 versus 9 days; and longer hospital stay, 34 versus 18 days. The frequency of nosocomial pneumonia was higher after unplanned extubation than after planned extubation: 28% versus 14%; this increase was entirely explained by accidental extubation. ICU mortality was 40.9% in patients with accidental extubation, 21.1% in patients with self-extubation, and 38% in the control group—these rates did not differ statistically.[8] In a retrospective case-control study, Epstein et al[65] also observed no difference in mortality between patients experiencing unplanned extubation, 32%, and a control group, 30%. The mortality in the control groups of both these series[8,71] is considerably higher than the mortality of less than 10% usually reported in successfully extubated patients.

Tominaga et al[72] undertook a prospective study of four interventions designed to influence the frequency of unplanned extubation. At baseline, 15.2% of intubated patients experienced unplanned extubation. When the investigators switched from their usual method of securing the endotracheal tube, cloth or Velcro ties, to the use of waterproof tape around the tube, upper lip, and face, the frequency of unplanned extubations decreased to 4% (8/213). The frequency was not altered by the more liberal use of sedative or paralytic agents. A decrease in the use of hand restraints led to an increase in the rate of unplanned extubations.

Thille et al[31] found that the distance between the endotracheal tube and the carina was significantly less in thirty-one patients who e [illegible] h in 168 instances of planned extubation: 4.1 ± 1.8 versus 5.6 ± 2.6 (SD). Given the high mortality rate in patients requiring reintubation, greater attention to the location of the tip of the endotracheal tube at the time of viewing chest radiographs could decrease this adverse event (Fig. 59-7).

Figure 59-7



Endotracheal tube positioned too high. Portable anteroposterior chest radiograph of a 24-year-old woman intubated as a result of anaphylactic shock. The endotracheal tube is 5.5 cm above the carina.

Conclusion

The development of severe respiratory distress after removal of an endotracheal tube, if sufficient to require reintubation, is associated with a high mortality rate. Clinicians are accordingly cautious and try to avoid premature extubation. Despite such caution, as many as two of every ten extubated patients require reintubation. To improve accuracy in forecasting extubation outcome, clinicians use diagnostic testing. The major diagnostic test for this purpose is a weaning trial. But unlike weaning predictor diagnostic tests, the diagnostic accuracy of weaning trials in predicting the outcome of a trial of extubation is unknown. Moreover, the accuracy is impossible to determine, because the experiments necessary to measure the sensitivity and specificity of a weaning trial (for predicting extubation outcome) are unethical. The mechanisms of weaning failure have been intensely investigated. Enhanced understanding of the pathophysiology has led to new approaches to the timing of the weaning process, prediction of outcome, and techniques used for weaning. In contrast, understanding of the pathophysiology of severe respiratory distress in the postextubation period is rudimentary to nonexistent. Acquiring such knowledge poses a great research challenge, but holds great promise for significantly advancing patient care.

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